Project Detail by Department

District	Project Type	CBS No	Project Name	Fund	$ in 1000's Fiscal Year 2017	2018-2022	6-Yr Total
Countywide	Sanitation	CBS-2722	Countywide Landfill PV/Wind Turbine Lighting	SW	195	0	195
		CBS-3182	Environmental Compliance System Design and	SW	250	7,750	8,000
	Sewer	CBS-1119	Countywide Environmental Protection Agency (EPA) Consent Decree Wastewater Reclamation Facility Renovation Projects	GB	0	7,500	7,500
				WF	1,600	2,500	4,100
		CBS-1128	Countywide Wastewater System Modifications	WF	500	5,500	6,000
		CBS-1132	Countywide Environmental Protection Agency (EPA) Compliance Projects	WF	500	2,500	3,000
		CBS-2732	Countywide Satellite Recycled Water Facilities	SRF	0	25,000	25,000
				WF	200	1,000	1,200
		CBS-2734	Countywide WWRF Chlorination System Upgrade	SRF	2,000	0	2,000
		CBS-3213	Countywide Permit Compliance Projects	WF	0	750	750
Hana	Drainage	CBS-2289	Hana Cinder Pit Grading and Closure Construction	SW	0	1,350	1,350
	Sanitation	CBS-1908	Hana Landfill Makai Berm Waste Removal	GB	2,000	2,000	4,000
Kihei-Makena	Sewer	CBS-1148	Kihei-Makena Environmental Protection Agency	WF	0	2,500	2,500
		CBS-1160	South Maui Recycled Water System Expansion	SRF	0	6,500	6,500
		CBS-1164	Maui Meadows Sewer System	WF	0	1,500	1,500
		CBS-1917	Kulanihakoi Street Recycled Water Line Extension	SRF	1,700	0	1,700
		CBS-2321	North Kihei Mauka Transmission System	SRF	0	15,000	15,000
				WF	200	1,500	1,700
		CBS-2322	Kihei No.7 Force Main Replacement	WF	0	1,500	1,500
		CBS-2729	Kihei No. 6 Force Main Replacement	WF	0	650	650
		CBS-2730	Kihei Wastewater Reclamation Facility RTU Upgrades	WF	1,500	0	1,500
		CBS-3196	Kenolio Road/Koki Place Sewer Rehabilitation	WF	50	400	450
		CBS-3197	Kihei No.8 Force Main Replacement	WF	0	200	200
		CBS-3198	Kihei Wastewater Pump Station No. 3 Modification/Upgrade	WF	0	200	200
		CBS-3199	Kihei Wastewater Pump Station No. 4 Modification/Upgrade	WF	0	200	200
		CBS-3200	South Kihei Road Gravity System Capacity Upgrade	SRF	0	4,000	4,000
				WF	0	500	500
Molokai	Sanitation	CBS-3184	Molokai Landfill - Environmental Assesment Amendment	SW	160	0	160
	Sewer	CBS-1153	Kaunakakai Effluent Force Main Replacement	SRF	0	1,900	1,900
				WF	0	200	200
Paia-Haiku		CBS-1183	Kuau No. 3 Force Main Replacement	WF	0	690	690
		CBS-1184	Kuau No. 4 Force Main Replacement	WF	0	600	600
		CBS-1955	Kuau No. 1 Force Main Replacement	WF	0	550	550
		CBS-1964	Kuau No. 2 Force Main Replacement	WF	0	550	550
		CBS-3207	Paia Wastewater Pump Station Modifications	WF	0	200	200
Wailuku-Kahului	Sanitation	CBS-1109	Central Maui Landfill (CML) Phase V Gas Collection System Expansion	GB	2,500	0	2,500
		CBS-1901	Central Maui Landfill Phases VI-B and VI-C Property	SW	0	1,000	1,000
		CBS-3175	Leachate Collection & Recovery and Electrical Distribution System Upgrades	SW	80	500	580
		CBS-3181	Central Maui Landfill System Control and Data Acquisition	SW	80	300	380

Project Detail by Department (Continued)

District	Project Type	CBS No	Project Name	Fund	$ in 1000's Fiscal Year 2017	2018-2022	6-Yr Total
	Sewer	CBS-1131	Environmental Protection Agency (EPA) Sewer	SRF	0	2,000	2,000
				WF	1,000	4,000	5,000
		CBS-1158	Waiko Road Subdivision Sewer System	WF	0	2,150	2,150
		CBS-1169	Wailuku-Kahului Wastewater Reclamation Facility	SRF	0	5,300	5,300
				WF	0	700	700
		CBS-1171	Wailuku-Kahului Recycled Water Force Main	SRF	0	6,200	6,200
				WF	0	500	500
		CBS-2323	Wailuku Wastewater Pump Station Modifications	WF	0	100	100
		CBS-3204	Upper Waiko Road Sewer Extension	SRF	0	2,000	2,000
				WF	0	150	150
		CBS-3205	Old Wailuku Heights Sewer System	WF	0	300	300
		CBS-3206	Central Maui Regional WWRF (Waikapu)	SRF	0	46,000	46,000
				WF	0	2,300	2,300
		CBS-3211	Kaa Wastewater Pump Station Modification	WF	0	200	200
		CBS-3212	Hoo Hui Ana Wastewater Pump Station Modifications	WF	0	100	100
		CBS-3233	Wailuku-Kahului WWRF Solids Building Renovation	SRF	1,000	0	1,000
West Maui	Government Facilities	CBS-3174	Lahaina Refuse Office	GB	250	0	250
	Sewer	CBS-1124	West Maui Recycled Water System Expansion	SRF	0	14,000	14,000
		CBS-1144	Lahaina Wastewater Reclamation Facility (WWRF)	SRF	12,500	9,000	21,500
		CBS-1146	Environmental Protection Agency (EPA) Sewer Rehabilitation	WF	0	2,000	2,000
		CBS-1161	Napili No. 3 Force Main Replacement	SRF	0	1,800	1,800
				WF	200	0	200
		CBS-1162	Napili Nos. 5 and 6 Force Main Replacements	WF	3,200	0	3,200
		CBS-1163	Sheraton Wastewater Pump Station Modifications	SRF	1,000	0	1,000
		CBS-1167	Napili No. 1 Force Main Replacement	SRF	0	2,700	2,700
				WF	0	300	300
		CBS-1168	Napili No. 2 Force Main Replacement	WF	0	1,100	1,100
		CBS-1174	Napili No. 4 Force Main Replacement	SRF	0	1,800	1,800
				WF	200	0	200
		CBS-1177	Napili Wastewater Pump Station No. 1 Modifications	SRF	0	6,800	6,800
				WF	0	630	630
		CBS-1178	Napili Wastewater Pump Station No. 2 Modifications	SRF	0	4,800	4,800
				WF	0	480	480
		CBS-1179	Napili Wastewater Pump Station No. 3 Modifications	SRF	0	4,200	4,200
				WF	0	400	400
		CBS-1180	Napili Wastewater Pump Station No. 4 Modifications	SRF	0	3,800	3,800
				WF	0	400	400
		CBS-1181	Napili Wastewater Pump Station No. 5 Modifications	SRF	0	2,300	2,300
		CBS-1182	Napili Wastewater Pump Station No. 6 Modifications	SRF	0	2,300	2,300
		CBS-1952	Kaanapali Resort R-1 Water Distribution System Expansion	SRF	0	7,000	7,000
		CBS-1954	Honokowai R-1 Water Distribution System	WF	0	2,200	2,200
		CBS-1968	Honoapiilani Highway Recycled Water Lateral Installation	WF	0	500	500

Project Detail by Department (Continued)

District	Project Type	CBS No	Project Name	Fund	$ in 1000's Fiscal Year 2017	2018-2022	6-Yr Total
		CBS-3201	Lahaina WWRF Onsite Recycled Water Storage	WF	0	150	150
		CBS-3202	Lahaina Wastewater Reclamation Facility Upgrade for Direct Potable Reuse	WF	150	0	150
			Total: Department of Environmental Management		**33,015**	**237,650**	**270,665**

FUNDING SOURCE SUMMARY	$ in 1000's Fiscal Year		
Fund	2017	2018-2022	6-Yr Total
AH	0	0	0
BW	0	0	0
FD	0	0	0
GB	4,750	9,500	14,250
GF	0	0	0
HF	0	0	0
LBF	0	0	0
LF	0	0	0
OG	0	0	0
OT	0	0	0
PA	0	0	0
RQ	0	0	0
SRF	18,200	174,400	192,600
ST	0	0	0
SW	765	10,900	11,665
US	0	0	0
WF	9,300	42,850	52,150
WR	0	0	0
WU	0	0	0
Total	**33,015**	**237,650**	**270,665**

NOTE: PROJECT SHEETS ARE INCLUDED ONLY FOR PROJECTS WITH FY 2017 FUNDING

CBS No: CBS-2722

Project Name: Countywide Landfill PV/Wind Turbine Lighting

Department: Department of Environmental Management

District: Countywide

Project Type: Sanitation

Anticipated Life: 7-years

Prior Years	Appr	Ensuing	Subsequent Years					Total
Expend/Encb	FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022	6-Year
0	0	195,000	0	0	0	0	0	195,000

PROJECT DESCRIPTION

This project will design and install renewable energy lighting systems for site safety and security at the County landfills. These PV/wind turbine powered lighting systems will be used at the landfills in areas where no electric power exists, to provide lighting to critical site access areas during low light conditions.

PROJECT JUSTIFICATION

The critical access areas of the County's landfills such as leachate wells, storm water detention basins, landfill gas wells, and landfill gas flares are located in remote areas where conventional electrical distribution does not exist, and access during low light conditions is a concern for the safety and security of landfill staff. The cost of providing alternative energy power to these systems is more favorable than providing the necessary infrastructure and hard wired electrical service for conventional lighting. This project will provide lighting for these sensitive and critical areas to allow employees to safely monitor and access during low light conditions in order to collect regulatory data, samples, and maintain/troubleshoot the system equipment.

STRATEGIC PLAN ALIGNMENT

Department's Strategic Plan	Countywide Priority Results
Safety and regulatory compliance.	A Suitable Public Infrastructure An Efficient, Effective, and Responsive Government A Prepared, Safe, and Liveable County A Healthy and Sustainable Community

Operating Impact Narrative

No impact is anticipated at this time.

FUNDING DETAILS

Phase Description	Fund Code	Appr	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022
New Construction	SW	0	195,000	0	0	0	0	0

Capital Improvement Program

Schedule of Activities			
Activity	**Start**	**End**	**Amount**
New Construction	10/01/2016	06/30/2017	195,000
Total Capital Project Costs			195,000
Total O&M Costs			35,000
Total Capital & Operating Costs			230,000

Methods of Financing (Ensuing + 5 Years)	
Funding Source	**Amount**
Solid Waste Fund	195,000
Total Funding Requirements	195,000

Capital Improvement Program

CBS No: CBS-3182

Project Name: Environmental Compliance System Design and Construction

Department: Department of Environmental Management

District: Countywide

Project Type: Sanitation

Anticipated Life: N/A

Prior Years	Appr	Ensuing	Subsequent Years					Total
Expend/Encb	FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022	6-Year
0	0	250,000	750,000	500,000	500,000	6,000,000	0	8,000,000

PROJECT DESCRIPTION

This project will provide design and construction services to address environmental compliance opportunities and operational improvements at all county solid waste management facilities in alignment with state and federal requirements.

PROJECT JUSTIFICATION

This project will allow the Solid Waste Division to address improvement opportunities at all county landfills and enable operational enhancements for environmental regulatory requirements. This includes mechanical, electrical, and civil design and construction services throughout the county.

STRATEGIC PLAN ALIGNMENT

Department's Strategic Plan	Countywide Priority Results
Enables the Division to quickly respond to emergency situations.	A Suitable Public Infrastructure A Strong, Diversified Economy An Efficient, Effective, and Responsive Government A Prepared, Safe, and Livable County A Healthy and Sustainable Community

OPERATING IMPACT NARRATIVE

The improvements made as a result of this project, will enable efficient and reliable monitoring and operation of county landfill environmental compliance systems.

FUNDING DETAILS

Phase Description	Fund Code	Appr FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022
Design	SW	0	100,000	300,000	200,000	200,000	1,000,000	0
New Construction	SW	0	150,000	450,000	300,000	300,000	5,000,000	0

Schedule of Activities

Activity	Start	End	Amount
Design	07/01/2017	12/31/2021	1,800,000
New Construction	07/01/2017	12/31/2018	6,200,000
Total Capital Project Costs			**8,000,000**
Total O&M Costs			0
Total Capital & Operating Costs			**8,000,000**

Methods of Financing (Ensuing + 5 Years)

Funding Source	Amount
Solid Waste Fund	8,000,000
Total Funding Requirements	8,000,000

Capital Improvement Program

CBS No: CBS-1119

Project Name: Countywide Environmental Protection Agency (EPA) Consent Decree Wastewater Reclamation Facility Renovation Projects

Department: Department of Environmental Management

District: Countywide

Project Type: Sewer

Anticipated Life: 20 years

Prior Years	Appr	Ensuing	Subsequent Years					Total
Expend/Encb	FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022	6-Year
2,243,483	2,000,000	1,600,000	2,000,000	2,000,000	2,000,000	2,000,000	2,000,000	11,600,000

PROJECT DESCRIPTION

The purpose of this project is to provide a program for design and construction, renovation or upgrade of equipment at the Wastewater Reclamation Facilities as required by the 20 Year Rehabilitation & Replacement Plan developed under the 1999 Consent Decree. Projects for FY2017 include: Electrical System upgrades at the Lahaina and Kahului WWRF's, repairs to headworks, chlorine mixing rooms and replacement of aeration discs at Kahului WWRF.

PROJECT JUSTIFICATION

Failure to provide for the programmed renovation, improvements & upgrade of all wastewater reclamation facilities/wastewater pump stations could result in continuous degradation and failures of equipment and process units. This might result in possible raw wastewater spills and additional fines from the EPA/DOH. Additionally, many of these projects will result in implementing energy saving designs and equipment.

STRATEGIC PLAN ALIGNMENT

Department's Strategic Plan	Countywide Priority Results
Sustain Reliable Wastewater Infrastructure Ensure Facilities Meet Future Needs Provide Reliable Wastewater Service	A Suitable Public Infrastructure A Strong, Diversified Economy An Efficient, Effective, and Responsive Government A Prepared, Safe, and Livable County A Healthy and Sustainable Community

OPERATING IMPACT NARRATIVE

No impact on staffing or operating budget anticipated.

FUNDING DETAILS

Phase Description	Fund Code	Appr FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022
New Construction	GB	0	0	1,500,000	1,500,000	1,500,000	1,500,000	1,500,000
New Construction	WF	2,000,000	1,600,000	500,000	500,000	500,000	500,000	500,000

Capital Improvement Program

Schedule of Activities			
Activity	**Start**	**End**	**Amount**
New Construction	07/01/2012	12/31/2021	11,600,000
Total Capital Project Costs			**11,600,000**
Total O&M Costs			0
Total Capital & Operating Costs			**11,600,000**

Methods of Financing (Ensuing + 5 Years)	
Funding Source	**Amount**
General Obligation Fund	7,500,000
Sewer Fund	4,100,000
Total Funding Requirements	**11,600,000**

Capital Improvement Program

CBS No: CBS-1128

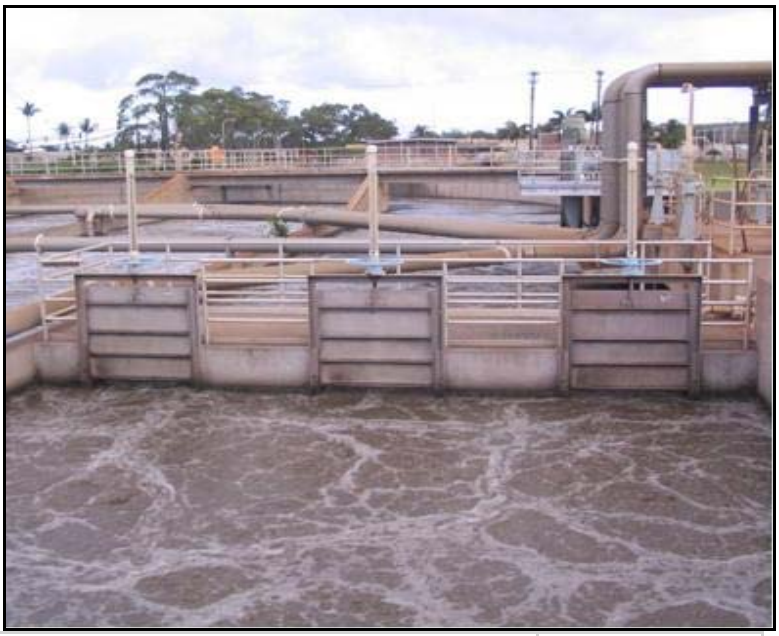
Project Name: Countywide Wastewater System Modifications

Department: Department of Environmental Management

District: Countywide

Project Type: Sewer

Anticipated Life: 20 years



Prior Years	Appr	Ensuing	Subsequent Years					Total
Expend/Encb	FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022	6-Year
4,289,613	500,000	500,000	1,500,000	1,000,000	1,000,000	1,000,000	1,000,000	6,000,000

PROJECT DESCRIPTION

Additional funding to be used only as required for projects where construction/design bids exceed the funding appropriation. Bid amount overruns have become common in recent years with the fluctuating construction climate in Maui County and global uncertainty of material and energy prices.

PROJECT JUSTIFICATION

The result of not establishing this project fund may be to postpone required infrastructure projects or to increase all projects estimates and lose the flexibility of placing funds where needed in order to accomplish the division's mission.

STRATEGIC PLAN ALIGNMENT

Department's Strategic Plan	Countywide Priority Results
Sustain Reliable Wastewater Infrastructure Ensure Facilities Meet Future Needs Provide Reliable Wastewater Service	A Suitable Public Infrastructure A Strong, Diversified Economy A Prepared, Safe, and Livable County A Healthy and Sustainable Community

OPERATING IMPACT NARRATIVE

No impact on staffing or operating budget anticipated.

FUNDING DETAILS

Phase Description	Fund Code	Appr FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022
New Construction	WF	500,000	1,000,000	1,000,000	1,000,000	1,000,000	1,000,000	1,000,000

Schedule of Activities

Activity	Start	End	Amount
New Construction	07/01/2012	12/31/2023	6,000,000
Total Capital Project Costs			**6,000,000**
Total O&M Costs			0
Total Capital & Operating Costs			**6,000,000**

Methods of Financing (Ensuing + 5 Years)

Funding Source	Amount
Sewer Fund	6,000,000
Total Funding Requirements	**6,000,000**

Capital Improvement Program

CBS No: CBS-1132

Project Name: Countywide Environmental Protection Agency (EPA) Compliance Projects

Department: Department of Environmental Management

District: Countywide

Project Type: Sewer

Anticipated Life: 20 years

Prior Years	Appr	Ensuing	Subsequent Years					Total
Expend/Encb	FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022	6-Year
3,481,635	500,000	500,000	500,000	500,000	500,000	500,000	500,000	3,000,000

PROJECT DESCRIPTION

The Environmental Protection Agency (EPA)/Department of Health (DOH) issued a Consent Decree (CD) dated November 8, 1999. The CD required the County of Maui to conduct numerous studies, investigations, and to prepare identified plans. Based on these documents, additional rehabilitation/construction projects are developed and executed. The exact scope of projects and additional assessments are not know at this time. A current CCTV project will identify needs when completed.

PROJECT JUSTIFICATION

Damaged or deteriorated lines require greater maintenance and pose the threat of potential blockages or total failure which may cause wastewater spills, damage to property and the environment, and increase operational costs due to callout of personnel. Scheduling the repair or replacement of deficiencies noted during recent evaluations is a requirement of the 1999 Consent Decree.

STRATEGIC PLAN ALIGNMENT

Department's Strategic Plan	Countywide Priority Results
Sustain Reliable Wastewater Infrastructure Ensure Facilities Meet Future Needs Provide Reliable Wastewater Service	A Suitable Public Infrastructure A Strong, Diversified Economy A Prepared, Safe, and Livable County A Healthy and Sustainable Community

OPERATING IMPACT NARRATIVE

No impact on staffing or operating budget anticipated.

FUNDING DETAILS

Phase Description	Fund Code	Appr FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022
New Construction	WF	500,000	500,000	500,000	500,000	500,000	500,000	500,000

Schedule of Activities

Activity	Start	End	Amount
New Construction	07/01/2012	12/31/2022	3,000,000
Total Capital Project Costs			**3,000,000**
			0
Total Capital & Operating Costs			**3,000,000**

Methods of Financing (Ensuing + 5 Years)

Funding Source	Amount
Sewer Fund	3,000,000
Total Funding Requirements	**3,000,000**

Capital Improvement Program

CBS No: CBS-2732

Project Name: Countywide Satellite Recycled Water Facilities

Department: Department of Environmental Management

District: Countywide

Project Type: Sewer

Anticipated Life: 30 years

Prior Years	Appr	Ensuing	Subsequent Years					Total
Expend/Encb	FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022	6-Year
0	0	200,000	0	1,000,000	0	12,500,000	12,500,000	26,200,000

PROJECT DESCRIPTION

Continued growth in all geographic sectors of the County is creating an opportunity to recycle water closer to the generation source. The FY2017 project will look at the feasibility and possible locations of the facilities necessary to achieve this goal and the following years for design and construction of facilities.

PROJECT JUSTIFICATION

Several factors are driving the need to look at the feasibility of this project: (1) This would maintain capacity in existing sewer collection lines and postpone replacement; (2) will promote recycled water usage in lieu of potable or brackish ground water; (3) lower distribution system costs of recycled water; (4) lessen the dependence on injection wells for final effluent discharge.

STRATEGIC PLAN ALIGNMENT

Department's Strategic Plan	Countywide Priority Results
Sustain Reliable Wastewater Infrastructure Ensure Facilities Meet Future Needs Provide Reliable Wastewater Service	A Suitable Public Infrastructure A Strong, Diversified Economy An Efficient, Effective, and Responsive Government A Healthy and Sustainable Community

OPERATING IMPACT NARRATIVE

No significant impact on staffing or operating budget is anticipated for this first phase.

FUNDING DETAILS

Phase Description	Fund Code	Appr FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022
Design	WF	0	0	0	1,000,000	0	0	0
New Construction	SRF	0	0	0	0	0	12,500,000	12,500,000
Planning	WF	0	200,000	0	0	0	0	0

Capital Improvement Program

Schedule of Activities			
Activity	**Start**	**End**	**Amount**
Design	11/01/2018	12/31/2019	1,000,000
New Construction	09/01/2020	12/31/2022	25,000,000
Planning	09/01/2016	06/30/2017	200,000
Total Capital Project Costs			**26,200,000**
Total O&M Costs			0
Total Capital & Operating Costs			**26,200,000**

Methods of Financing (Ensuing + 5 Years)	
Funding Source	**Amount**
Sewer Fund	1,200,000
State Revolving Loan Fund	25,000,000
Total Funding Requirements	**26,200,000**

CBS No: CBS-2734

Project Name: Countywide WWRF Chlorination System Upgrade

Department: Department of Environmental Management

District: Countywide

Project Type: Sewer

Anticipated Life: 50 years

Prior Years	Appr	Ensuing	Subsequent Years					Total
Expend/Encb	FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022	6-Year
0	0	2,000,000	0	0	0	0	0	2,000,000

PROJECT DESCRIPTION

Upgrade of an aging and vulnerable chlorination system required for algae control and effluent disinfection.

PROJECT JUSTIFICATION

The current system is approaching the end of its useful life and an upgrade is required to keep staff safe and maintain quality water processing.

STRATEGIC PLAN ALIGNMENT

Department's Strategic Plan	Countywide Priority Results
Sustain Reliable Wastewater Infrastructure Ensure Facilities Meet Future Needs Provide Reliable Wastewater Service	A Suitable Public Infrastructure A Strong, Diversified Economy An Efficient, Effective, and Responsive Government A Prepared, Safe, and Liveable County A Healthy and Sustainable Community

Operating Impact Narrative

No significant impact on staffing or operating budget is anticipated.

FUNDING DETAILS

Phase Description	Fund Code	Appr	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022
New Construction	SRF	0	2,000,000	0	0	0	0	0

Schedule of Activities

Activity	Start	End	Amount
New Construction	07/01/2016	12/31/2017	2,000,000
Total Capital Project Costs			2,000,000
Total O&M Costs			0
Total Capital & Operating Costs			2,000,000

Methods of Financing (Ensuing + 5 Years)

Funding Source	Amount
State Revolving Loan Fund	2,000,000
Total Funding Requirements	2,000,000

Capital Improvement Program

CBS No: CBS-1908

Project Name: Hana Landfill Makai Berm Waste Removal

Department: Department of Environmental Management

District: Hana

Project Type: Sanitation

Anticipated Life: 20 Years

Prior Years	Appr	Ensuing	Subsequent Years					Total
Expend/Encb	FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022	6-Year
901,494	250,000	2,000,000	2,000,000	0	0	0	0	4,000,000

PROJECT DESCRIPTION

This project requests funds to remediate un-permitted waste dumped outside of the existing permitted waste limits of Hana Landfill, as a result of the Hawaii Department of Health requirements for solid waste management. This multi-year phased approach will achieve clean closure by removing the waste from the Makai side of the landfill, and relocating the waste to the current permitted landfill parcel in alignment with the HDOH remediation requirements.

PROJECT JUSTIFICATION

During a recent site inspection, Hawaii Department of Health noted to the County of the potential impending violation for un-permitted dumping of waste outside permitted limits at the Hana Landfill. HDOH quoted specifically the state and federal regulations against "constructing and operating a waste disposal unit outside the 'existing' limits of waste, without complying with the design criteria for municipal solid waste landfills, and without a solid waste management permit." This project will resolve this deficiency, by relocating the waste to the proper permitted landfill area.

STRATEGIC PLAN ALIGNMENT

Department's Strategic Plan	Countywide Priority Results
Environmental and regulatory compliance. Promotes environmental sustainability.	A Suitable Public Infrastructure An Efficient, Effective, and Responsive Government A Prepared, Safe, and Livable County A Healthy and Sustainable Community

OPERATING IMPACT NARRATIVE

Separating soil will provide for landfill cover savings. Staff will need to accommodate and coordinate with onsite contractor(s) during project duration.

FUNDING DETAILS

Phase Description	Fund Code	Appr FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022
Renovations	SW	250,000	0	0	0	0	0	0
Renovations	GB	0	2,000,000	2,000,000	0	0	0	0

Capital Improvement Program

Schedule of Activities			
Activity	**Start**	**End**	**Amount**
Renovations	07/01/2016	12/31/2018	4,000,000
Total Capital Project Costs			**4,000,000**
Total O&M Costs			0
Total Capital & Operating Costs			**4,000,000**

Methods of Financing (Ensuing + 5 Years)	
Funding Source	**Amount**
General Obligation Fund	4,000,000
Total Funding Requirements	**4,000,000**

Capital Improvement Program

CBS No: CBS-1917

Project Name: Kulanihakoi Street Recycled Water Line Extension

Department: Department of Environmental Management

District: Kihei-Makena

Project Type: Sewer

Anticipated Life: 40 years

Prior Years	Appr	Ensuing	Subsequent Years					Total
Expend/Encb	FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022	6-Year
0	0	1,700,000	0	0	0	0	0	1,700,000

PROJECT DESCRIPTION

Design and construction of an extension to the Kihei recycled water distribution system. This new 2,200 lineal foot pipeline will deliver recycled water for irrigation use to the planned Kihei High School and other multifamily and commercial customers.

PROJECT JUSTIFICATION

Expansion of the recycled water distribution system will reduce the demands of potable water in the South Maui area and prevent over pumping of local brackish water reserves. The design will allow for a future loop connection along the North-South collector road to provide increased reliability to customers.

STRATEGIC PLAN ALIGNMENT

Department's Strategic Plan	Countywide Priority Results
Sustain Reliable Wastewater Infrastructure Ensure Facilities Meet Future Needs Provide Reliable Wastewater Service	A Suitable Public Infrastructure A Strong, Diversified Economy A Prepared, Safe, and Livable County A Healthy and Sustainable Community

OPERATING IMPACT NARRATIVE

No significant impact on staffing or operating budget is anticipated.

FUNDING DETAILS

Phase Description	Fund Code	Appr FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022
New Construction	SRF	0	1,700,000	0	0	0	0	0

Schedule of Activities

Activity	Start	End	Amount
New Construction	09/01/2016	12/01/2017	1,700,000
Total Capital Project Costs			**1,700,000**
Total O&M Costs			0
Total Capital & Operating Costs			**1,700,000**

Methods of Financing (Ensuing + 5 Years)

Funding Source	Amount
State Revolving Loan Fund	1,700,000
Total Funding Requirements	**1,700,000**

Capital Improvement Program

CBS No: CBS-2321

Project Name: North Kihei Mauka Transmission System

Department: Department of Environmental Management

District: Kihei-Makena

Project Type: Sewer

Anticipated Life: 30 years

Prior Years	Appr	Ensuing	Subsequent Years					Total
Expend/Encb	FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022	6-Year
0	0	200,000	0	1,500,000	0	15,000,000	0	16,700,000

PROJECT DESCRIPTION

Preliminary Study for the design and construction of a wastewater collection and transmission system for development mauka of Piilani Highway.

PROJECT JUSTIFICATION

Sewer capacity in the main gravity collection lines and force mains within South Kihei Road are beginning to reach capacity. Any future development mauka of Piilani Highway will require a separate system to address the capacity issues and reduce the potential effects of sewer spills into the ocean.

STRATEGIC PLAN ALIGNMENT

Department's Strategic Plan	Countywide Priority Results
Sustain Reliable Wastewater Infrastructure Ensure Facilities Meet Future Needs Provide Reliable Wastewater Service	A Suitable Public Infrastructure A Strong, Diversified Economy A Prepared, Safe, and Livable County A Healthy and Sustainable Community

OPERATING IMPACT NARRATIVE

Will require additional resources in man hours, equipment maintenance and power. No long term staffing impacts are anticipated.

FUNDING DETAILS

Phase Description	Fund Code	Appr FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022
Design	WF	0	0	0	1,500,000	0	0	0
New Construction	SRF	0	0	0	0	0	15,000,000	0
Planning	WF	0	200,000	0	0	0	0	0

Capital Improvement Program

Schedule of Activities			
Activity	**Start**	**End**	**Amount**
Design	09/01/2018	12/31/2019	1,500,000
New Construction	09/01/2020	12/31/2021	15,000,000
Planning	09/01/2016	12/31/2017	200,000
Total Capital Project Costs			**16,700,000**
Total O&M Costs			0
Total Capital & Operating Costs			**16,700,000**

Methods of Financing (Ensuing + 5 Years)	
Funding Source	**Amount**
Sewer Fund	1,700,000
State Revolving Loan Fund	15,000,000
Total Funding Requirements	**16,700,000**

Capital Improvement Program

CBS No: CBS-2730

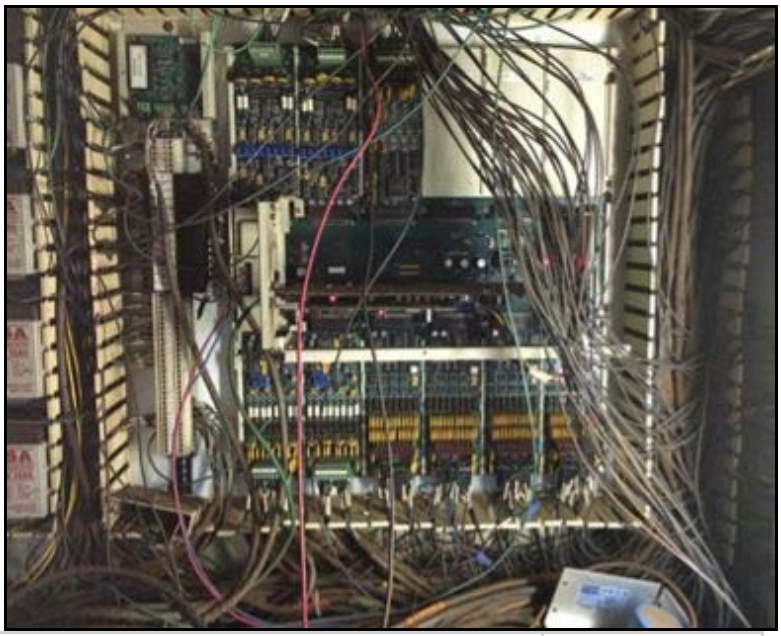
Project Name: Kihei Wastewater Reclamation Facility RTU Upgrades

Department: Department of Environmental Management

District: Kihei-Makena

Project Type: Sewer

Anticipated Life: 20 years



Prior Years	Appr	Ensuing	Subsequent Years					Total
Expend/Encb	FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022	6-Year
0	0	1,500,000	0	0	0	0	0	1,500,000

PROJECT DESCRIPTION

Replacement of twelve radio transmitting units and the associated programming to make the new units function properly. This is a complicated project because it involves a high level of programming and the documentation of the existing system is minimal. The design will require that all points in the system be tested and mapped or that the system be rebuilt completely.

PROJECT JUSTIFICATION

The RTUs at Kihei WWRF have reached the end of their useful lives. The current units are obsolete and we cannot get the repair parts needed to fix them or replacements should they go down. Additionally, all spare parts have been used. If any unit fails staff will be required to man the plant 24 hours per day because the transmitters relay critical information to the SCADA system and auto-dialer which notifies personnel of an emergency situation.

STRATEGIC PLAN ALIGNMENT

Department's Strategic Plan	Countywide Priority Results
Sustain Reliable Wastewater Infrastructure Ensure Facilities Meet Future Needs Provide Reliable Wastewater Service	A Suitable Public Infrastructure An Efficient, Effective, and Responsive Government A Prepared, Safe, and Livable County A Healthy and Sustainable Community

OPERATING IMPACT NARRATIVE

No significant impact on staffing or operating budget is anticipated.

FUNDING DETAILS

Phase Description	Fund Code	Appr FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022
New Construction	WF	0	1,500,000	0	0	0	0	0

Schedule of Activities

Activity	Start	End	Amount
New Construction	09/01/2016	12/31/2017	1,500,000
Total Capital Project Costs			**1,500,000**
Total O&M Costs			0
Total Capital & Operating Costs			**1,500,000**

Methods of Financing (Ensuing + 5 Years)

Funding Source	Amount
Sewer Fund	1,500,000
Total Funding Requirements	**1,500,000**

Capital Improvement Program

CBS No: CBS-3196

Project Name: Kenolio Road/Koki Place Sewer Rehabilitation

Department: Department of Environmental Management

District: Kihei-Makena

Project Type: Sewer

Anticipated Life: 50 Years

Prior Years	Appr	Ensuing	Subsequent Years					Total
Expend/Encb	FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022	6-Year
0	0	50,000	0	400,000	0	0	0	450,000

PROJECT DESCRIPTION

Replacement of existing sewer line in disrepair and responsible for several sewage spills in the last 10 years. Actual replacement will be determined upon completion of current CCTV investigation.

PROJECT JUSTIFICATION

Site is in an easement area with difficult access issues. Currently staff is required to flush lines regularly in an attempt to avoid blockages due to debris buildup. Replacement will reduce maintenance costs and spill potential.

STRATEGIC PLAN ALIGNMENT

Department's Strategic Plan	Countywide Priority Results
Sustain Reliable Wastewater Infrastructure Ensure Facilities Meet Future Needs Provide Reliable Wastewater Service	A Suitable Public Infrastructure An Efficient, Effective, and Responsive Government A Prepared, Safe, and livable County A Healthy and Sustainable Community

OPERATING IMPACT NARRATIVE

No impact on staffing but a reduction in cost due to the elimination of callouts for spills (that typically require overtime payment)

FUNDING DETAILS

Phase Description	Fund Code	Appr FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022
Design	WF	0	50,000	0	0	0	0	0
New Construction	WF	0	0	0	400,000	0	0	0

Schedule of Activities

Activity	Start	End	Amount
Design	09/01/2016	12/31/2017	50,000
New Construction	09/01/2018	12/31/2019	400,000
Total Capital Project Costs			**450,000**
Total O&M Costs			0
Total Capital & Operating Costs			**450,000**

Methods of Financing (Ensuing + 5 Years)

Funding Source	Amount
Sewer Fund	450,000
Total Funding Requirements	**450,000**

Capital Improvement Program

CBS No: CBS-3184

Project Name: Molokai Landfill - Environmental Assessment Amendment

Department: Department of Environmental Management

District: Molokai

Project Type: Sanitation

Anticipated Life: 20 Years

Prior Years	Appr	Ensuing	Subsequent Years					Total
Expend/Encb	FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022	6-Year
0	0	160,000	0	0	0	0	0	160,000

PROJECT DESCRIPTION

This project will perform an Environmental Assessment (EA) amendment to include provisions for future phases 5 & 6 bringing the proposed landfill footprint to approximately 19-acres extending the life of the landfill by approximately 26 years.

PROJECT JUSTIFICATION

The original landfill area extents of the permitted 12 acres under the existing EA have been reached with the expansion of the Molokai landfill Phase 4. The EA needs to be amended to be inclusive of the entire landfill area as capacity within the current limits is projected to be exhausted in 2020 (permitting could take up to 24 months). This EA amendment is the first critical path task needed to expand the landfill operation into phases 5 & 6, in order to ensure uninterrupted solid waste management services for the island of Molokai.

STRATEGIC PLAN ALIGNMENT

Department's Strategic Plan	Countywide Priority Results
This project is the first step in providing the necessary infrastructure to support a healthy and sustainable community.	A Suitable Public Infrastructure An Efficient, Effective, and Responsive Government A Prepared, Safe, and Livable County A Healthy and Sustainable Community

OPERATING IMPACT NARRATIVE

No significant impact on staffing or operating budget is anticipated.

FUNDING DETAILS

Phase Description	Fund Code	Appr FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022
Planning	SW	0	160,000	0	0	0	0	0

Schedule of Activities

Activity	Start	End	Amount
Planning	07/01/2016	12/31/2017	160,000
Total Capital Project Costs			**160,000**
Total O&M Costs			0
Total Capital & Operating Costs			**160,000**

Methods of Financing (Ensuing + 5 Years)

Funding Source	Amount
Solid Waste Fund	160,000
Total Funding Requirements	**160,000**

Capital Improvement Program

CBS No: CBS-1109

Project Name: Central Maui Landfill (CML) Phase V Gas Collection System Expansion

Department: Department of Environmental Management

District: Wailuku-Kahului

Project Type: Sanitation

Anticipated Life: 20 years

Prior Years	Appr	Ensuing	Subsequent Years					Total
Expend/Encb	FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022	6-Year
637,200	0	2,500,000	0	0	0	0	0	2,500,000

PROJECT DESCRIPTION

This project will provide the necessary infrastructure expansion to collect and control landfill gas at Central Maui Landfill Phase V as per state and federal regulations.

PROJECT JUSTIFICATION

In accordance with federal and state regulations, landfill gas must be collected within 5 years of landfilling in an expansion area. Phase V started to receive municipal waste in 2011, and thus the deadline for the completion of the gas collection system for the entire Phase V permitted area will required to be completed by 2016. This project will complete the required construction and infrastructure to meet the requirements of gas collection as a condition of the County's covered source permit, and will be completed by Year End 2016. Failure to construct the complete system, and thus comply with this requirement in FY 2017, will subject the County Solid Waste Division to fines and penalties by both federal and state agencies.

STRATEGIC PLAN ALIGNMENT

Department's Strategic Plan	Countywide Priority Results
Project will help the Division to continue managing solid waste in accordance with its operating and NSPS permit requirements. Regulatory compliance is a goal of the Division's planning document, the ISWMP.	A Suitable Public Infrastructure An Efficient, Effective, and Responsive Government A Prepared, Safe, and Livable County A Healthy and Sustainable Community

OPERATING IMPACT NARRATIVE

With the expansion of the gas collection system, additional landfill gas wellheads, piping, and ancillary equipment will need to be monitored, adjusted, and periodically maintained in working order for the life of the permitted landfill area.

FUNDING DETAILS

Phase Description	Fund Code	Appr FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022
New Construction	GB	0	2,500,000	0	0	0	0	0

Capital Improvement Program

Schedule of Activities			
Activity	**Start**	**End**	**Amount**
New Construction	11/01/2017	12/31/2018	2,500,000
Total Capital Project Costs			**2,500,000**
Total O&M Costs			105,000
Total Capital & Operating Costs			**2,605,000**

Methods of Financing (Ensuing + 5 Years)	
Funding Source	**Amount**
General Obligation Fund	2,500,000
Total Funding Requirements	**2,500,000**

Capital Improvement Program

CBS No: CBS-3175

Project Name: Leachate Collection & Recovery and Electrical Distribution System Upgrades

Department: Department of Environmental Management

District: Wailuku-Kahului

Project Type: Sanitation

Anticipated Life: 20 Years

Prior Years	Appr	Ensuing	Subsequent Years					Total
Expend/Encb	FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022	6-Year
0	0	80,000	500,000	0	0	0	0	580,000

PROJECT DESCRIPTION

This project will upgrade and modernize the leachate collection and recovery system at the Central Maui Landfill Phase IV-A Wet Well and Leachate Storage Tanks. The scope of this project includes upgrading the electrical distribution system to meet current NEC requirements, necessary mechanical design upgrades, along with providing reliable and un-interrupted operation of this Mission critical control system required by the site solid waste permit.

PROJECT JUSTIFICATION

The LCRS rehabilitation shall improve secondary containment systems and monitoring of leachate quantities as required by the State of Hawaii Department of Health (DOH). Current underground structure has received a notice of violation and fine due to the inabilities to manage leachate quantities effectively in 2011. This project will improve site safety by reducing the frequency of confined space entries; provide leachate measurement and alarm notification to operations to assist with regulatory documentation requirements; and mechanical design upgrades to ensure that there is no lapse in the system operation.

STRATEGIC PLAN ALIGNMENT

Department's Strategic Plan	Countywide Priority Results
This project will provide necessary infrastructure to meet the community's environmentally responsible expectations and demands.	A Suitable Public Infrastructure An Efficient, Effective, and Responsive Government A Prepared, Safe, and Livable County A Healthy and Sustainable Community

OPERATING IMPACT NARRATIVE

The proposed project shall provide the proper secondary containment of leachate, increase the accuracy and efficiency of monitoring the leachate quantities, and improve overall system safety for facility personnel. This project will require periodic maintenance for the mechanical systems to ensure reliable and safe operation of the equipment.

FUNDING DETAILS

Phase Description	Fund Code	Appr FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022
Design	SW	0	80,000	0	0	0	0	0
New Construction	SW	0	0	500,000	0	0	0	0

Capital Improvement Program

Schedule of Activities			
Activity	**Start**	**End**	**Amount**
Design	07/01/2016	12/01/2016	80,000
New Construction	12/01/2016	07/01/2017	500,000
Total Capital Project Costs			**580,000**
Total O&M Costs			11,000
Total Capital & Operating Costs			**591,000**

Methods of Financing (Ensuing + 5 Years)	
Funding Source	**Amount**
Solid Waste Fund	580,000
Total Funding Requirements	**580,000**

Capital Improvement Program

CBS No: CBS-3181

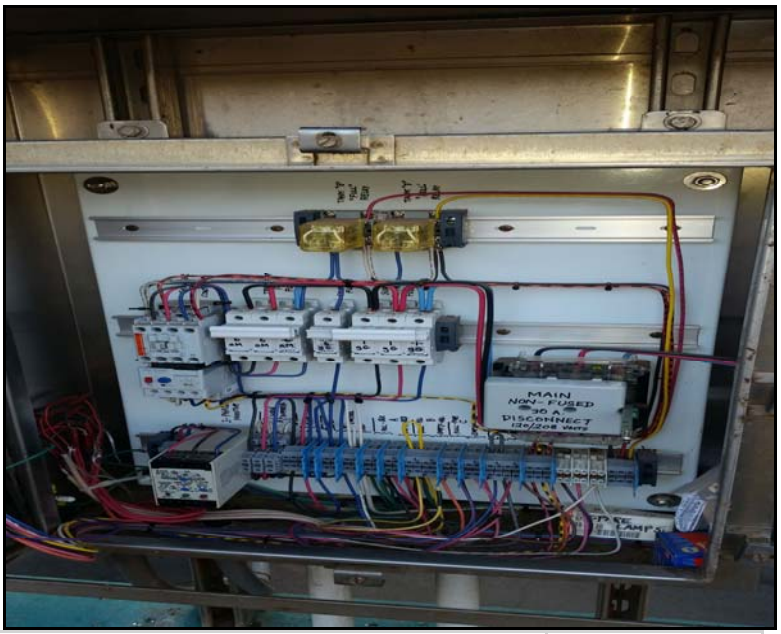
Project Name: Central Maui Landfill System Control and Data Acquisition

Department: Department of Environmental Management

District: Wailuku-Kahului

Project Type: Sanitation

Anticipated Life: 10 years



Prior Years	Appr	Ensuing	Subsequent Years					Total
Expend/Encb	FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022	6-Year
0	0	80,000	300,000	0	0	0	0	380,000

PROJECT DESCRIPTION

This project will provide the necessary infrastructure and equipment to monitor and record data in real time critical regulatory monitoring systems such as landfill gas, leachate collection, storm water, and meteorological data on-site and remotely for both operations and engineering sections for compliance requirements at the Central Maui Landfill. This project will bring regulatory monitoring to current best practices readily available in today's industry.

PROJECT JUSTIFICATION

This system will decrease risk of possible permit violations and EPA consent decrees, up to $25,000 per day per occurrence. The system telemetry and infrastructure provided through this project would enable operations, as well as engineering, to remotely monitor and record data needed per DOH regulatory requirements. The system would provide real time visualization and recording of equipment operation status, fault notifications, cumulative data collection, and other operational parameters in order to comply with site permit requirements and operational limits. These systems would be able to be monitored remotely, in contrast to the current need of having landfill and engineering staff inspections of individual system components spread out across the 150-acre landfill facility. Future expansion for additional monitoring through future projects will be provided as well as a result of this project.

STRATEGIC PLAN ALIGNMENT

Department's Strategic Plan	Countywide Priority Results
This project will provide the infrastructure to support operational and environmental sustainability.	A Suitable Public Infrastructure An Efficient, Effective, and Responsive Government A Prepared, Safe, and Livable County A Healthy and Sustainable Community

OPERATING IMPACT NARRATIVE

This project shall provide a reliable means to remotely monitor landfill systems according to the requirements of DOH. It shall also reduce the time required for operational staff to conduct daily recordkeeping and allow engineering to remotely troubleshoot systems.

FUNDING DETAILS

Phase Description	Fund Code	Appr FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022
Design	SW	0	80,000	0	0	0	0	0
New Construction	SW	0	0	300,000	0	0	0	0

Capital Improvement Program

Schedule of Activities			
Activity	**Start**	**End**	**Amount**
Design	07/01/2016	12/31/2016	80,000
New Construction	02/01/2017	12/31/2017	300,000
Total Capital Project Costs			**380,000**
Total O&M Costs			6,000
Total Capital & Operating Costs			**386,000**

Methods of Financing (Ensuing + 5 Years)	
Funding Source	**Amount**
Solid Waste Fund	380,000
Total Funding Requirements	**380,000**

Capital Improvement Program

CBS No: CBS-1131

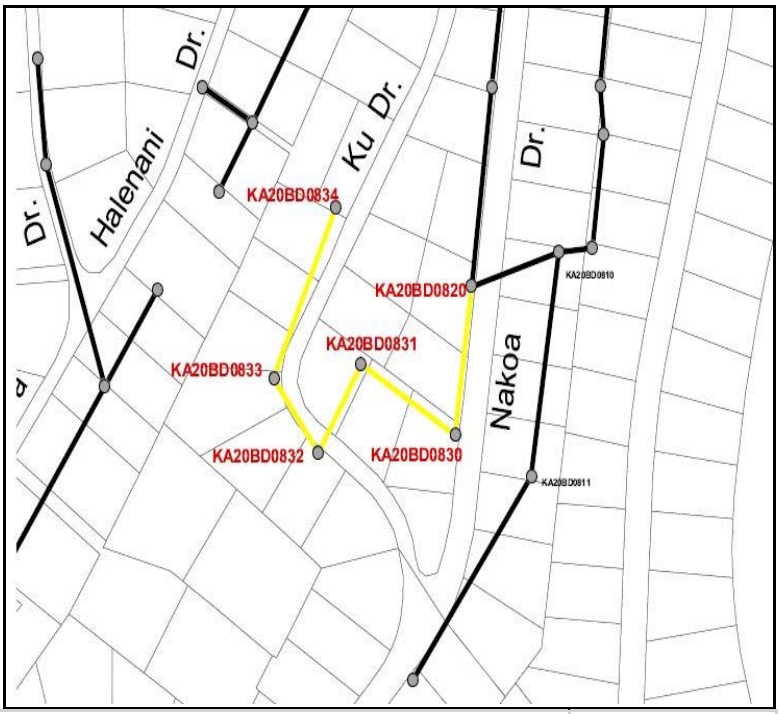
Project Name: Environmental Protection Agency (EPA) Sewer Rehabilitation

Department: Department of Environmental Management

District: Wailuku-Kahului

Project Type: Sewer

Anticipated Life: 50 years



Prior Years	Appr	Ensuing	Subsequent Years					Total
Expend/Encb	FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022	6-Year
609,103	750,000	1,000,000	1,000,000	2,000,000	0	2,000,000	1,000,000	7,000,000

PROJECT DESCRIPTION

Rehabilitation of major gravity sewer lines that have reached the end of their useful life and/or have been found to be damaged or in disrepair via video investigation or other means. Evaluation and project definitions are ongoing. FY2017 Scope: Construct Ku Drive Line Replacement, Design South and North Mokuhau Piihana Line replacements.

PROJECT JUSTIFICATION

Damaged or deteriorated lines require greater maintenance and pose the threat of potential blockages or total failure which may cause wastewater spills, damage to property and the environment, and increase operational costs due to callout of personnel. Spills would entail fines established by the Consent Decree. Scheduling the repair or replacement of deficiencies noted during recent evaluations is a requirement of the 1999 Consent Decree.

STRATEGIC PLAN ALIGNMENT

Department's Strategic Plan	Countywide Priority Results
Sustain Reliable Wastewater Infrastructure Ensure Facilities Meet Future Needs Provide Reliable Wastewater Service	A Suitable Public Infrastructure A Strong, Diversified Economy A Prepared, Safe, and Livable County A Healthy and Sustainable Community

OPERATING IMPACT NARRATIVE

No impact on staffing or operating budget anticipated.

FUNDING DETAILS

Phase Description	Fund Code	Appr FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022
Renovations	SRF	0	0	0	0	0	2,000,000	0
Renovations	WF	750,000	1,000,000	1,000,000	2,000,000	0	0	1,000,000

Capital Improvement Program

Schedule of Activities			
Activity	**Start**	**End**	**Amount**
Renovations	09/12/2012	12/31/2022	7,000,000
Total Capital Project Costs			**7,000,000**
Total O&M Costs			0
Total Capital & Operating Costs			**7,**000,000

Methods of Financing (Ensuing + 5 Years)	
Funding Source	**Amount**
Sewer Fund	5,000,000
State Revolving Loan Fund	2,000,000
Total Funding Requirements	**7,000,000**

Capital Improvement Program

CBS No: CBS-3233

Project Name: Wailuku-Kahului WWRF Solids Building Renovation

Department: Department of Environmental Management

District: Wailuku-Kahului

Project Type: Sewer

Anticipated Life: 20 years

Prior Years	Appr	Ensuing	Subsequent Years					Total
Expend/Encb	FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022	6-Year
0	0	1,000,000	0	0	0	0	0	1,000,000

PROJECT DESCRIPTION

Renovations to the Solids Handling Building located at the Wailuku-Kahului Wastewater Reclamation Facility (WWRF). Scope of work includes structural, electrical and cosmetic aspects to the interior and exterior of the structure.

PROJECT JUSTIFICATION

The building is located adjacent to the beach area and is subject to constant salt water corrosion and strong winds. Corrosion of structural beams, columns and building siding and roofing has caused substantial leakage and interior/exterior damage. This building houses critical equipment for separating solid from liquid waste, as well as, supplies that require proper protection from the elements.

STRATEGIC PLAN ALIGNMENT

Department's Strategic Plan	Countywide Priority Results
Sustain Reliable Wastewater Infrastructure Ensure Facilities Meet Future Needs Provide Reliable Wastewater Service	A Suitable Public Infrastructure An Efficient, Effective, and Responsive Government A Prepared, Safe, and Livable County A Healthy and Sustainable Community

OPERATING IMPACT NARRATIVE

No significant impact on staffing or operating budget is anticipated.

FUNDING DETAILS

Phase Description	Fund Code	Appr FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022
Renovations	SRF	0	1,000,000	0	0	0	0	0

Schedule of Activities

Activity	Start	End	Amount
Renovations	01/01/2017	12/31/2017	1,000,000
Total Capital Project Costs			**1,000,000**
Total O&M Costs			0
Total Capital & Operating Costs			**1,000,000**

Methods of Financing (Ensuing + 5 Years)

Funding Source	Amount
State Revolving Loan Fund	1,000,000
Total Funding Requirements	1,000,000

Capital Improvement Program

CBS No: CBS-3174

Project Name: Lahaina Refuse Office

Department: Department of Environmental Management

District: West Maui

Project Type: Government Facilities

Anticipated Life: 20 years

Prior Years	Appr	Ensuing	Subsequent Years					Total
Expend/Encb	FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022	6-Year
0	0	250,000	0	0	0	0	0	250,000

PROJECT DESCRIPTION

This project will provide a new mobile office for the Lahaina refuse collection employees to better meet the requirements of the operational and personnel growth in the district.

PROJECT JUSTIFICATION

The existing office for the Lahaina Refuse Collection Staff is in immediate need of replacement due to having reached the end of its useful life with current damage and repairs beyond its current value. Multiple movements of the trailer from its former site at the Lahaina WWRF have caused several unsuitable occupancy and safety concerns. The replacement of this office will provide the Lahaina Refuse employees the proper office space, infrastructure, proper utilities, and sanitary needs to ensure the safe and productive operation of the refuse collection team.

STRATEGIC PLAN ALIGNMENT

Department's Strategic Plan	Countywide Priority Results
Plan and implement the necessary infrastructure investments to support a healthy and sustainable community.	A Suitable Public Infrastructure An Efficient, Effective, and Responsive Government A Prepared, Safe, and Livable County A Healthy and Sustainable Community

OPERATING IMPACT NARRATIVE

Refuse collection crew will have ample space to utilize for daily operations, training, and meetings.

FUNDING DETAILS

Phase Description	Fund Code	Appr FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022
Other/Manufacture/Installation	GB	0	250,000	0	0	0	0	0

Schedule of Activities

Activity	Start	End	Amount
Other	07/01/2016	06/30/2017	250,000
Total Capital Project Costs			**250,000**
Total O&M Costs			30,000
Total Capital & Operating Costs			280,000

Methods of Financing (Ensuing + 5 Years)

Funding Source	Amount
General Obligation Fund	250,000
Total Funding Requirements	**250,000**

Capital Improvement Program

CBS No: CBS-1144

Project Name: Lahaina Wastewater Reclamation Facility (WWRF) Modifications, Stage IA

Department: Department of Environmental Management

District: West Maui

Project Type: Sewer

Anticipated Life: 20 years

Prior Years	Appr	Ensuing	Subsequent Years					Total
Expend/Encb	FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022	6-Year
1,000,000	12,500,000	12,500,000	9,000,000	0	0	0	0	21,500,000

PROJECT DESCRIPTION

Provide engineering consultant services to design long term operational process changes to the 1975 and 1985 treatment facilities; provide complete design plans and bid documents for required work; and complete construction of basins and other required facilities.

PROJECT JUSTIFICATION

Due to the age of the facility and changing permit discharge requirements, the current process operations cannot meet the reserve allocations required for Kapalua, HFDC, Kaanapali properties and other proposed projects. Major modifications and upgrades need to be completed in order to have the reserve capacity available for development. This upgrade will allow required maintenance activities to occur such as cleaning clarifiers or aeration basins and will restore the 9.0 mgd capacity at the site.

STRATEGIC PLAN ALIGNMENT

Department's Strategic Plan	Countywide Priority Results
Sustain Reliable Wastewater Infrastructure Ensure Facilities Meet Future Needs Provide Reliable Wastewater Service	A Suitable Public Infrastructure A Strong, Diversified Economy An Efficient, Effective, and Responsive Government A Prepared, Safe, and Livable County A Healthy and Sustainable Community

OPERATING IMPACT NARRATIVE

Anticipated increase in electrical costs and maintenance as new basins/clarifies or other structures are constructed and utilized. Unable to provide O&M breakdown.

FUNDING DETAILS

Phase Description	Fund Code	Appr FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022
New Construction	SRF	12,500,000	12,500,000	9,000,000	0	0	0	0

Capital Improvement Program

Schedule of Activities			
Activity	**Start**	**End**	**Amount**
New Construction	07/01/2015	12/31/2018	21,500,000
Total Capital Project Costs			**21,500,000**
Total O&M Costs			0
Total Capital & Operating Costs			**21,500,000**

Methods of Financing (Ensuing + 5 Years)	
Funding Source	**Amount**
State Revolving Loan Fund	21,500,000
Total Funding Requirements	**21,500,000**

Capital Improvement Program

CBS No: CBS-1161

Project Name: Napili No. 3 Force Main Replacement

Department: Department of Environmental Management

District: West Maui

Project Type: Sewer

Anticipated Life: 30 years

Prior Years	Appr	Ensuing	Subsequent Years					Total
Expend/Encb	FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022	6-Year
0	0	200,000	0	1,800,000	0	0	0	2,000,000

PROJECT DESCRIPTION

Replacement of a 2,074 foot long, 16 inch diameter ductile iron pipe force main constructed in 1985. This line is within Lower Honoapiilani Road (between Hui Nui Place to Hoohui Road) and is nearing the end of its useful life.

PROJECT JUSTIFICATION

A 2004 corrosion study of County owned force mains identified the condition and estimated remaining useful life of these assets. Based on this information the Wastewater Reclamation Division developed a planned replacement schedule for its aging facilities. This replacement is necessary in order to avoid a costly failure of the line and possible major sewage spill to the adjacent park, beach and ocean, as well as, fines from the Environmental Protection Agency (EPA) as stipulated in the Consent Decree.

STRATEGIC PLAN ALIGNMENT

Department's Strategic Plan	Countywide Priority Results
Sustain Reliable Wastewater Infrastructure Ensure Facilities Meet Future Needs Provide Reliable Wastewater Service	A Suitable Public Infrastructure A Strong, Diversified Economy A Prepared, Safe, and Livable County A Healthy and Sustainable Community

OPERATING IMPACT NARRATIVE

No significant long term impact on staffing or operating budget is anticipated.

FUNDING DETAILS

Phase Description	Fund Code	Appr FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022
Design	WF	0	200,000	0	0	0	0	0
New Construction	SRF	0	0	0	1,800,000	0	0	0

Capital Improvement Program

Schedule of Activities			
Activity	**Start**	**End**	**Amount**
Design	09/01/2016	12/31/2017	200,000
New Construction	09/01/2018	12/31/2019	1,800,000
Total Capital Project Costs			**2,000,000**
Total O&M Costs			0
Total Capital & Operating Costs			**2,000,000**

Methods of Financing (Ensuing + 5 Years)	
Funding Source	**Amount**
Sewer Fund	200,000
State Revolving Loan Fund	1,800,000
Total Funding Requirements	**2,000,000**

Capital Improvement Program

CBS No: CBS-1162

Project Name: Napili No. 5 and 6 Force Main Replacements

Department: Department of Environmental Management

District: West Maui

Project Type: Sewer

Anticipated Life: 30 years

Prior Years	Appr	Ensuing	Subsequent Years					Total
Expend/Encb	FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022	6-Year
300,000	0	3,200,000	0	0	0	0	0	3,200,000

PROJECT DESCRIPTION

Replacement of two interconnected 14-inch diameter ductile iron pipe force main within Lower Honoapiilani Road which were constructed in 1985 and are nearing the end of their useful life.

PROJECT JUSTIFICATION

A 2004 corrosion study of County owned force mains identified the condition of several force mains and estimated remaining useful life. Based on this information the Wastewater Reclamation Division developed a planned replacement schedule for its aging facilities. This replacement is necessary in order to avoid a costly failure of the line and possible major sewage spill to the adjacent park, beach and ocean, as well as, fines from the Environmental Protection Agency (EPA) as stipulated in the Consent Decree.

STRATEGIC PLAN ALIGNMENT

Department's Strategic Plan	Countywide Priority Results
Sustain Reliable Wastewater Infrastructure Ensure Facilities Meet Future Needs Provide Reliable Wastewater Service	A Suitable Public Infrastructure A Strong, Diversified Economy A Prepared, Safe, and Livable County A Healthy and Sustainable Community

OPERATING IMPACT NARRATIVE

No significant long term impact on staffing or operating budget is anticipated.

FUNDING DETAILS

Phase Description	Fund Code	Appr FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022
New Construction	WF	0	3,200,000	0	0	0	0	0

Capital Improvement Program

Schedule of Activities			
Activity	**Start**	**End**	**Amount**
New Construction	09/01/2016	12/31/2017	3,200,000
Total Capital Project Costs			**3,200,000**
Total O&M Costs			0
Total Capital & Operating Costs			**3,200,000**

Methods of Financing (Ensuing + 5 Years)	
Funding Source	**Amount**
Sewer Fund	3,200,000
Total Funding Requirements	**3,200,000**

Capital Improvement Program

CBS No: CBS-1163

Project Name: Sheraton Wastewater Pump Station Modifications

Department: Department of Environmental Management

District: West Maui

Project Type: Sewer

Anticipated Life: 20 years

Prior Years	Appr	Ensuing	Subsequent Years					Total
Expend/Encb	FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022	6-Year
25,000	80,000	1,000,000	0	0	0	0	0	1,000,000

PROJECT DESCRIPTION

The purpose of this project is to design and construct upgrades (pumps, motors, generators, SCADA, etc.) and required modifications (building, piping, electrical) to the existing pump station located on the Kaanapali Golf course and originally constructed in 1967. This station needs to meet area capacity requirements and be standardized to be similar in design and operation as other West Maui pump station facilities.

PROJECT JUSTIFICATION

This facility is reaching the end of its useful life. Acquiring critical spare parts has become difficult and the reliability of the equipment is beginning to be in question. Modifications are required to maintain reliable service, reduce maintenance costs and prevent possible sewage spills. Failure of the pump station would result is wastewater spills directly onto the golf course and into storm drains and the ocean.

STRATEGIC PLAN ALIGNMENT

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OPERATING IMPACT NARRATIVE

No significant impact on staffing or operating budget is anticipated.

FUNDING DETAILS

Phase Description	Fund Code	Appr FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022
Design	WF	80,000	0	0	0	0	0	0
New Construction	SRF	0	1,000,000	0	0	0	0	0

Capital Improvement Program

Schedule of Activities			
Activity	**Start**	**End**	**Amount**
Design	10/01/2015	10/31/2016	0
New Construction	12/01/2016	12/31/2017	1,000,000
Total Capital Project Costs			**1,000,000**
Total O&M Costs			0
Total Capital & Operating Costs			**1,000,000**

Methods of Financing (Ensuing + 5 Years)	
Funding Source	**Amount**
State Revolving Loan Fund	1,000,000
Total Funding Requirements	**1,000,000**

Capital Improvement Program

CBS No: CBS-1174

Project Name: Napili No. 4 Force Main Replacement

Department: Department of Environmental Management

District: West Maui

Project Type: Sewer

Anticipated Life: 30 years

Prior Years	Appr	Ensuing	Subsequent Years					Total
Expend/Encb	FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022	6-Year
0	0	200,000	0	1,800,000	0	0	0	2,000,000

PROJECT DESCRIPTION

Replacement of a 2,040 foot long, 14" diameter ductile iron pipe force main within Lower Honoapiilani Road (between Hui Road "F" and Napilihau St.) which was constructed in 1985. This pipeline is nearing the end of its useful life.

PROJECT JUSTIFICATION

A 2004 corrosion study of County owned force mains identified the condition and estimated remaining useful life of these assets. Based on this information the Wastewater Reclamation Division developed a planned replacement schedule for its aging facilities. This replacement is necessary in order to avoid a costly failure of the line and possible major sewage spill to the adjacent homes, beach, drainage courses and ocean, as well as, fines from the Environmental Protection Agency (EPA) as stipulated in the Consent Decree.

STRATEGIC PLAN ALIGNMENT

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OPERATING IMPACT NARRATIVE

No significant impact on staffing or operating budget is anticipated.

FUNDING DETAILS

Phase Description	Fund Code	Appr FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022
Design	WF	0	200,000	0	0	0	0	0
New Construction	SRF	0	0	0	1,800,000	0	0	0

Capital Improvement Program

Schedule of Activities			
Activity	**Start**	**End**	**Amount**
Design	09/01/2016	12/31/2019	200,000
New Construction	10/01/2018	12/01/2019	1,800,000
Total Capital Project Costs			**2,000,000**
Total O&M Costs			0
Total Capital & Operating Costs			**2,000,000**

Methods of Financing (Ensuing + 5 Years)	
Funding Source	**Amount**
Sewer Fund	200,000
State Revolving Loan Fund	1,800,000
Total Funding Requirements	**2,000,000**

Capital Improvement Program

CBS No: CBS-3202

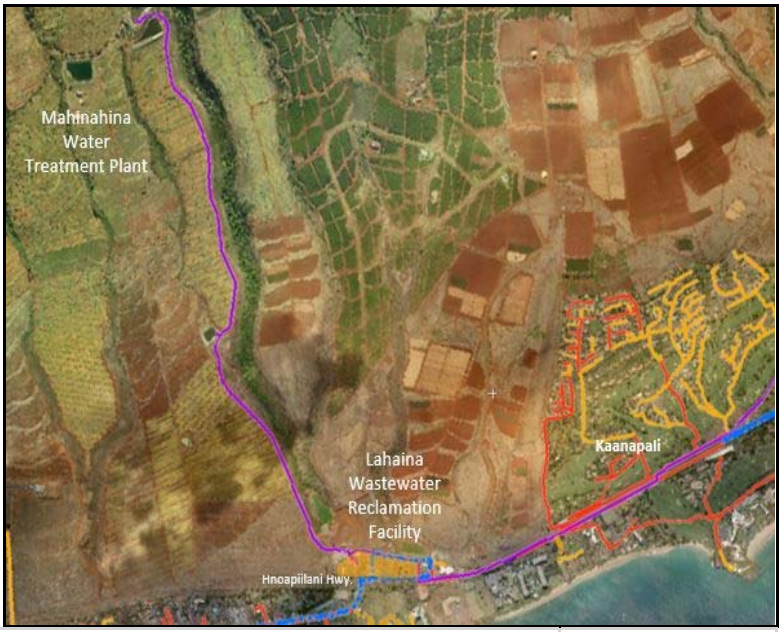
Project Name: Lahaina Wastewater Reclamation Facility Upgrade for Direct Potable Reuse

Department: Department of Environmental Management

District: West Maui

Project Type: Sewer

Anticipated Life: 20 years



Prior Years	Appr	Ensuing	Subsequent Years					Total
Expend/Encb	FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022	6-Year
0	0	150,000	0	0	0	0	0	150,000

PROJECT DESCRIPTION

Project Engineering Report (PER) to determine the requirements of additional processes required to refine treated wastewater to a quality suitable for introduction to the County's potable water supply.

PROJECT JUSTIFICATION

Continuing pressure by certain community members and organizations to increase the volume of reuse of treated wastewater and decrease/eliminate the use of injection wells makes this a primary candidate for reuse as is currently being demonstrated by several municipalities across the United States.

STRATEGIC PLAN ALIGNMENT

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OPERATING IMPACT NARRATIVE

The PER will have no impact. If the project proceeds, the PER will estimate the capital and future costs to the community.

FUNDING DETAILS

Phase Description	Fund Code	Appr FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	FY 2021	FY 2022
Planning	WF	0	150,000	0	0	0	0	0

Schedule of Activities

Activity	Start	End	Amount
Planning	07/01/2016	12/31/2017	150,000
Total Capital Project Costs			**150,000**
Total O&M Costs			0
Total Capital & Operating Costs			**150,000**

Methods of Financing (Ensuing + 5 Years)

Funding Source	Amount
Sewer Fund	150,000
Total Funding Requirements	**150,000**